VersaPlanetary Load Ratings Guide

Introduction

The ratings in this guide are based on actual test data conducted by VEX Robotics on the final product versions of the VersaPlanetary gear box. All load ratings are based on a Safety Factor (SF) of 1.2 to accommodate manufacturing tolerance differences.

How to Use the Rating Tables

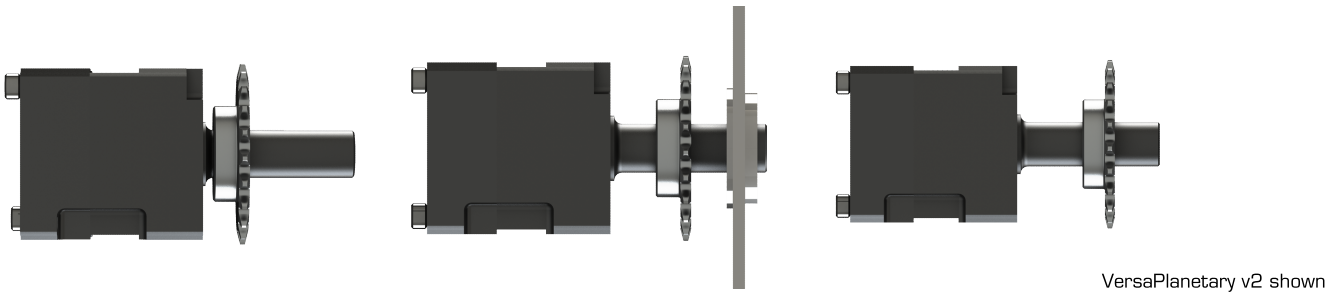
There are two types of rating tables in this guide:

(1) Simple-Load Ratings
   a. Conservative ratings based on a simplified loading case using just torsion
(2) Combined-Load Ratings
   a. Load ratings that take bending loads into account as well as torsion

It is recommended to mount the VersaPlanetary gearboxes such that "Simple Load Ratings" may be used.

Important Note:

The "Simple Load Ratings" tables assume that the output of the shaft has minimal overhung loading (i.e. the sprocket is mounted close to the base of the shaft, or the tip of the shaft is supported). See below examples.



VersaPlanetary v2 shown

Sprocket close to transmission

Use Simple Load Ratings

Sprocket tip supported

Use Simple Load Ratings

Sprocket NOT close to transmission and sprocket tip is unsupported

Use Combined Load Ratings

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Since the 7:1, 9:1, and 10:1 gear sets have a lower torque capacity that the 3:1, 4:1, 5:1 gear sets, it's possible for a VersaPlanetary built in one order to be weaker than a VersaPlanetary built in a different order. Ideally, the higher the reduction, the closer to the motor the gearset should go.

Example:
A VersaPlanetary that uses a 5:1 and 10:1 gear set to make a 50:1 gear reduction.

Motor	First Stage	Second Stage	Output Shaft	Torque Capacity
Motor	5:1	10:1	1/2" Hex	100 Nm
Motor	10:1	5:1	1/2" Hex	154 Nm

The first VersaPlanetary has the 10:1 gear set is placed as the final stage. In this situation, the 10:1 gear set is the limiting factor. The second VersaPlanetary has the 10:1 gear set placed as the first stage. In this situation, the 1/2" hex shaft is the limiting factor.

This means that it's possible for a 50:1 reduction to be both green and red depending on the order of the gear sets. When looking at the Simple Load Ratings, pay attention to the order of the gear sets.

VersaPlanetary 90° Drive
The VersaPlanetary 90° Drive is designed to withstand all of the recommended (green) ratios of the VersaPlanetary. As long as the VersaPlanetary around the 90° Drive uses a recommended ration, users won't experience any failures.

VersaPlanetary 180° Drive
The VersaPlanetary 180° Drive must be the first stage after the motor and is designed to handle the stall torque of the motors listed in this document. However, users should factor in any ratio used between the motor and the VersaPlanetary 180° Drive when reviewing the Simple Load Ratings tables.

VersaPlanetary Ratchet Slice
The VersaPlanetary Ratchet Slice will work anywhere in the VersaPlanetary stack. There are some considerations. For example, the VersaPlanetary Ratchet Slice will high greater resolution as it gets closer to the motor. In addition, the closer to the motor the ratchet slice is located, the less force it must hold in order to lock rotation. The failure mode observed during testing is that the mounting screw holes will fail first. This failure was observed at around 160 Nm of torque.

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Understanding The Simple Load Ratings

The “Simple Load Ratings” tables use a red/green rating system. If the motor/gear ratio/output shaft combination selected is highlighted green, then it is within the maximum gear ratio recommendations. If the motor/gear ratio/output shaft combination is highlighted red, then the maximum motor torque will be capable of damaging the gearbox (with a significant enough load on the output shaft) and should be used with caution.

Final gear ratio is indicated in the colored cells.

2-Stage Example:
User wants to use a Mini CIM motor with a 2 stage gearbox with a 100:1 gear ratio and a 3/8” Hex Shaft. Is this combination recommended?

Using table below, the “Stage 1” column is the stage closest to the motor and the “Stage 2” column is the gear set closest to the output shaft. As you can see, 100:1 is in red, this means that it is not recommended. However, any gear ratio less than 40:1 is recommended.

		Stage 2					
Motor	**Stage 1**	3:1	4:1	5:1	7:1	9:1	10:1
Mini CIM	3:1	9	12	15	21	27	30
	4:1	12	16	20	28	36	40
	5:1	15	20	25	35	45	50
	7:1	21	28	35	49	63	70
	9:1	27	36	45	63	81	90
	10:1	30	40	50	70	90	100

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3-Stage Example:
User wants to use a 775pro motor with a 3 stage gearbox with a 200:1 gear ratio and a 1/2" Hex Shaft. Is this combination recommended?

Using the table below, the "Stage 1 & 2" column is the combined ratio of the two stages closest to the motor. The "Stage 3" column is the gear set closest to the output shaft.

As you can see, 200:1 is in both red and green, this means the user needs to pay attention to the order of the gear ratios. For example, it would not be recommended to build a 20:1 and 10:1 VersaPlanetary. However, a 40:1 and a 5:1 would be fine.

		Stage 3					
Motor	**Stage 1 & 2**	3:1	4:1	5:1	7:1	9:1	10:1
775pro	9:1	27	36	45	63	81	90
	12:1	36	48	60	84	108	120
	15:1	45	60	75	105	135	150
	16:1	48	64	80	112	144	160
	20:1	60	80	100	140	180	200
	21:1	63	84	105	147	189	210
	25:1	75	100	125	175	225	250
	27:1	81	108	135	189	243	270
	28:1	84	112	140	196	252	280
	30:1	90	120	150	210	270	300
	35:1	105	140	175	245	315	350
	36:1	108	144	180	252	324	360
	40:1	120	160	200	280	360	400
	45:1	135	180	225	315	405	450
	49:1	147	196	245	343	441	490
	50:1	150	200	250	350	450	500
	63:1	189	252	315	441	567	630
	70:1	210	280	350	490	630	700
	81:1	243	324	405	567	729	810
	90:1	270	360	450	630	810	900
	100:1	300	400	500	700	900	1000

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Simple Load Ratings Tables

Table 1 - 2 Stage VersaPlanetary with 1/2" Hex Output Shaft

Motor	Stage 1	Stage 2					
		3:1	4:1	5:1	7:1	9:1	10:1
AM-9015	3:1	9	12	15	21	27	30
	4:1	12	16	20	28	36	40
	5:1	15	20	25	35	45	50
	7:1	21	28	35	49	63	70
	9:1	27	36	45	63	81	90
	10:1	30	40	50	70	90	100
BB RS-550	3:1	9	12	15	21	27	30
	4:1	12	16	20	28	36	40
	5:1	15	20	25	35	45	50
	7:1	21	28	35	49	63	70
	9:1	27	36	45	63	81	90
	10:1	30	40	50	70	90	100
BB RS-775	3:1	9	12	15	21	27	30
	4:1	12	16	20	28	36	40
	5:1	15	20	25	35	45	50
	7:1	21	28	35	49	63	70
	9:1	27	36	45	63	81	90
	10:1	30	40	50	70	90	100
BAG Motor	3:1	9	12	15	21	27	30
	4:1	12	16	20	28	36	40
	5:1	15	20	25	35	45	50
	7:1	21	28	35	49	63	70
	9:1	27	36	45	63	81	90
	10:1	30	40	50	70	90	100
775pro	3:1	9	12	15	21	27	30
	4:1	12	16	20	28	36	40
	5:1	15	20	25	35	45	50
	7:1	21	28	35	49	63	70
	9:1	27	36	45	63	81	90
	10:1	30	40	50	70	90	100

Motor	Stage 1	Stage 2					
		3:1	4:1	5:1	7:1	9:1	10:1
Mini CIM	3:1	9	12	15	21	27	30
	4:1	12	16	20	28	36	40
	5:1	15	20	25	35	45	50
	7:1	21	28	35	49	63	70
	9:1	27	36	45	63	81	90
	10:1	30	40	50	70	90	100
CIM	3:1	9	12	15	21	27	30
	4:1	12	16	20	28	36	40
	5:1	15	20	25	35	45	50
	7:1	21	28	35	49	63	70
	9:1	27	36	45	63	81	90
	10:1	30	40	50	70	90	100
NEO	3:1	9	12	15	21	27	30
	4:1	12	16	20	28	36	40
	5:1	15	20	25	35	45	50
	7:1	21	28	35	49	63	70
	9:1	27	36	45	63	81	90
	10:1	30	40	50	70	90	100
Falcon 500	3:1	9	12	15	21	27	30
	4:1	12	16	20	28	36	40
	5:1	15	20	25	35	45	50
	7:1	21	28	35	49	63	70
	9:1	27	36	45	63	81	90
	10:1	30	40	50	70	90	100

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Table 1 - 2 Stage VersaPlanetary with 3/8" Hex Output Shaft

Motor	Stage 1	Stage 2					
		3:1	4:1	5:1	7:1	9:1	10:1
AM-9015	3:1	9	12	15	21	27	30
	4:1	12	16	20	28	36	40
	5:1	15	20	25	35	45	50
	7:1	21	28	35	49	63	70
	9:1	27	36	45	63	81	90
	10:1	30	40	50	70	90	100
BB RS-550	3:1	9	12	15	21	27	30
	4:1	12	16	20	28	36	40
	5:1	15	20	25	35	45	50
	7:1	21	28	35	49	63	70
	9:1	27	36	45	63	81	90
	10:1	30	40	50	70	90	100
BB RS-775	3:1	9	12	15	21	27	30
	4:1	12	16	20	28	36	40
	5:1	15	20	25	35	45	50
	7:1	21	28	35	49	63	70
	9:1	27	36	45	63	81	90
	10:1	30	40	50	70	90	100
BAG Motor	3:1	9	12	15	21	27	30
	4:1	12	16	20	28	36	40
	5:1	15	20	25	35	45	50
	7:1	21	28	35	49	63	70
	9:1	27	36	45	63	81	90
	10:1	30	40	50	70	90	100
775pro	3:1	9	12	15	21	27	30
	4:1	12	16	20	28	36	40
	5:1	15	20	25	35	45	50
	7:1	21	28	35	49	63	70
	9:1	27	36	45	63	81	90
	10:1	30	40	50	70	90	100

Motor	Stage 1	Stage 2					
		3:1	4:1	5:1	7:1	9:1	10:1
Mini CIM	3:1	9	12	15	21	27	30
	4:1	12	16	20	28	36	40
	5:1	15	20	25	35	45	50
	7:1	21	28	35	49	63	70
	9:1	27	36	45	63	81	90
	10:1	30	40	50	70	90	100
CIM	3:1	9	12	15	21	27	30
	4:1	12	16	20	28	36	40
	5:1	15	20	25	35	45	50
	7:1	21	28	35	49	63	70
	9:1	27	36	45	63	81	90
	10:1	30	40	50	70	90	100
NEO	3:1	9	12	15	21	27	30
	4:1	12	16	20	28	36	40
	5:1	15	20	25	35	45	50
	7:1	21	28	35	49	63	70
	9:1	27	36	45	63	81	90
	10:1	30	40	50	70	90	100
Falcon 500	3:1	9	12	15	21	27	30
	4:1	12	16	20	28	36	40
	5:1	15	20	25	35	45	50
	7:1	21	28	35	49	63	70
	9:1	27	36	45	63	81	90
	10:1	30	40	50	70	90	100

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Table 1 - 3 Stage VersaPlanetary with 1/2" Hex Output Shaft

Motor	Stage 1 & 2	Stage 3					
		3:1	4:1	5:1	7:1	9:1	10:1
AM-9015	9:1	27	36	45	63	81	90
	12:1	36	48	60	84	108	120
	15:1	45	60	75	105	135	150
	16:1	48	64	80	112	144	160
	20:1	60	80	100	140	180	200
	21:1	63	84	105	147	189	210
	25:1	75	100	125	175	225	250
	27:1	81	108	135	189	243	270
	28:1	84	112	140	196	252	280
	30:1	90	120	150	210	270	300
	35:1	105	140	175	245	315	350
	36:1	108	144	180	252	324	360
	40:1	120	160	200	280	360	400
	45:1	135	180	225	315	405	450
	49:1	147	196	245	343	441	490
	50:1	150	200	250	350	450	500
	63:1	189	252	315	441	567	630
	70:1	210	280	350	490	630	700
	81:1	243	324	405	567	729	810
	90:1	270	360	450	630	810	900
	100:1	300	400	500	700	900	1000
BB RS-550	9:1	27	36	45	63	81	90
	12:1	36	48	60	84	108	120
	15:1	45	60	75	105	135	150
	16:1	48	64	80	112	144	160
	20:1	60	80	100	140	180	200
	21:1	63	84	105	147	189	210
	25:1	75	100	125	175	225	250
	27:1	81	108	135	189	243	270
	28:1	84	112	140	196	252	280
	30:1	90	120	150	210	270	300
	35:1	105	140	175	245	315	350
	36:1	108	144	180	252	324	360
	40:1	120	160	200	280	360	400
	45:1	135	180	225	315	405	450
	49:1	147	196	245	343	441	490
	50:1	150	200	250	350	450	500
	63:1	189	252	315	441	567	630
	70:1	210	280	350	490	630	700
	81:1	243	324	405	567	729	810
	90:1	270	360	450	630	810	900
	100:1	300	400	500	700	900	1000

Motor	Stage 1 & 2	Stage 3					
		3:1	4:1	5:1	7:1	9:1	10:1
BB RS-775	9:1	27	36	45	63	81	90
	12:1	36	48	60	84	108	120
	15:1	45	60	75	105	135	150
	16:1	48	64	80	112	144	160
	20:1	60	80	100	140	180	200
	21:1	63	84	105	147	189	210
	25:1	75	100	125	175	225	250
	27:1	81	108	135	189	243	270
	28:1	84	112	140	196	252	280
	30:1	90	120	150	210	270	300
	35:1	105	140	175	245	315	350
	36:1	108	144	180	252	324	360
	40:1	120	160	200	280	360	400
	45:1	135	180	225	315	405	450
	49:1	147	196	245	343	441	490
	50:1	150	200	250	350	450	500
	63:1	189	252	315	441	567	630
	70:1	210	280	350	490	630	700
	81:1	243	324	405	567	729	810
	90:1	270	360	450	630	810	900
	100:1	300	400	500	700	900	1000
BAG Motor	9:1	27	36	45	63	81	90
	12:1	36	48	60	84	108	120
	15:1	45	60	75	105	135	150
	16:1	48	64	80	112	144	160
	20:1	60	80	100	140	180	200
	21:1	63	84	105	147	189	210
	25:1	75	100	125	175	225	250
	27:1	81	108	135	189	243	270
	28:1	84	112	140	196	252	280
	30:1	90	120	150	210	270	300
	35:1	105	140	175	245	315	350
	36:1	108	144	180	252	324	360
	40:1	120	160	200	280	360	400
	45:1	135	180	225	315	405	450
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	70:1	210	280	350	490	630	700
	81:1	243	324	405	567	729	810
	90:1	270	360	450	630	810	900
	100:1	300	400	500	700	900	1000

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VersaPlanetary Load Ratings Guide

Table 1 - 3 Stage VersaPlanetary with 1/2" Hex Output Shaft (Cont.)

Motor	Stage 1 & 2	Stage 3					
		3:1	4:1	5:1	7:1	9:1	10:1
775pro	9:1	27	36	45	63	81	90
	12:1	36	48	60	84	108	120
	15:1	45	60	75	105	135	150
	16:1	48	64	80	112	144	160
	20:1	60	80	100	140	180	200
	21:1	63	84	105	147	189	210
	25:1	75	100	125	175	225	250
	27:1	81	108	135	189	243	270
	28:1	84	112	140	196	252	280
	30:1	90	120	150	210	270	300
	35:1	105	140	175	245	315	350
	36:1	108	144	180	252	324	360
	40:1	120	160	200	280	360	400
	45:1	135	180	225	315	405	450
	49:1	147	196	245	343	441	490
	50:1	150	200	250	350	450	500
	63:1	189	252	315	441	567	630
	70:1	210	280	350	490	630	700
	81:1	243	324	405	567	729	810
	90:1	270	360	450	630	810	900
	100:1	300	400	500	700	900	1000
Mini CIM	9:1	27	36	45	63	81	90
	12:1	36	48	60	84	108	120
	15:1	45	60	75	105	135	150
	16:1	48	64	80	112	144	160
	20:1	60	80	100	140	180	200
	21:1	63	84	105	147	189	210
	25:1	75	100	125	175	225	250
	27:1	81	108	135	189	243	270
	28:1	84	112	140	196	252	280
	30:1	90	120	150	210	270	300
	35:1	105	140	175	245	315	350
	36:1	108	144	180	252	324	360
	40:1	120	160	200	280	360	400
	45:1	135	180	225	315	405	450
	49:1	147	196	245	343	441	490
	50:1	150	200	250	350	450	500
	63:1	189	252	315	441	567	630
	70:1	210	280	350	490	630	700
	81:1	243	324	405	567	729	810
	90:1	270	360	450	630	810	900
	100:1	300	400	500	700	900	1000

Motor	Stage 1 & 2	Stage 3					
		3:1	4:1	5:1	7:1	9:1	10:1
CIM	9:1	27	36	45	63	81	90
	12:1	36	48	60	84	108	120
	15:1	45	60	75	105	135	150
	16:1	48	64	80	112	144	160
	20:1	60	80	100	140	180	200
	21:1	63	84	105	147	189	210
	25:1	75	100	125	175	225	250
	27:1	81	108	135	189	243	270
	28:1	84	112	140	196	252	280
	30:1	90	120	150	210	270	300
	35:1	105	140	175	245	315	350
	36:1	108	144	180	252	324	360
	40:1	120	160	200	280	360	400
	45:1	135	180	225	315	405	450
	49:1	147	196	245	343	441	490
	50:1	150	200	250	350	450	500
	63:1	189	252	315	441	567	630
	70:1	210	280	350	490	630	700
	81:1	243	324	405	567	729	810
	90:1	270	360	450	630	810	900
	100:1	300	400	500	700	900	1000
NEO	9:1	27	36	45	63	81	90
	12:1	36	48	60	84	108	120
	15:1	45	60	75	105	135	150
	16:1	48	64	80	112	144	160
	20:1	60	80	100	140	180	200
	21:1	63	84	105	147	189	210
	25:1	75	100	125	175	225	250
	27:1	81	108	135	189	243	270
	28:1	84	112	140	196	252	280
	30:1	90	120	150	210	270	300
	35:1	105	140	175	245	315	350
	36:1	108	144	180	252	324	360
	40:1	120	160	200	280	360	400
	45:1	135	180	225	315	405	450
	49:1	147	196	245	343	441	490
	50:1	150	200	250	350	450	500
	63:1	189	252	315	441	567	630
	70:1	210	280	350	490	630	700
	81:1	243	324	405	567	729	810
	90:1	270	360	450	630	810	900
	100:1	300	400	500	700	900	1000

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Table 1 - 3 Stage VersaPlanetary with 1/2" Hex Output Shaft (Cont.)

		Stage 3					
Motor	**Stage 1 & 2**	3:1	4:1	5:1	7:1	9:1	10:1
Falcon 500	9:1	27	36	45	63	81	90
	12:1	36	48	60	84	108	120
	15:1	45	60	75	105	135	150
	16:1	48	64	80	112	144	160
	20:1	60	80	100	140	180	200
	21:1	63	84	105	147	189	210
	25:1	75	100	125	175	225	250
	27:1	81	108	135	189	243	270
	28:1	84	112	140	196	252	280
	30:1	90	120	150	210	270	300
	35:1	105	140	175	245	315	350
	36:1	108	144	180	252	324	360
	40:1	120	160	200	280	360	400
	45:1	135	180	225	315	405	450
	49:1	147	196	245	343	441	490
	50:1	150	200	250	350	450	500
	63:1	189	252	315	441	567	630
	70:1	210	280	350	490	630	700
	81:1	243	324	405	567	729	810
	90:1	270	360	450	630	810	900
	100:1	300	400	500	700	900	1000

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Table 1 - 3 Stage VersaPlanetary with 3/8" Hex Output Shaft

Motor	Stage 1 & 2	Stage 3					
		3:1	4:1	5:1	7:1	9:1	10:1
AM-9015	9:1	27	36	45	63	81	90
	12:1	36	48	60	84	108	120
	15:1	45	60	75	105	135	150
	16:1	48	64	80	112	144	160
	20:1	60	80	100	140	180	200
	21:1	63	84	105	147	189	210
	25:1	75	100	125	175	225	250
	27:1	81	108	135	189	243	270
	28:1	84	112	140	196	252	280
	30:1	90	120	150	210	270	300
	35:1	105	140	175	245	315	350
	36:1	108	144	180	252	324	360
	40:1	120	160	200	280	360	400
	45:1	135	180	225	315	405	450
	49:1	147	196	245	343	441	490
	50:1	150	200	250	350	450	500
	63:1	189	252	315	441	567	630
	70:1	210	280	350	490	630	700
	81:1	243	324	405	567	729	810
	90:1	270	360	450	630	810	900
	100:1	300	400	500	700	900	1000
BB RS-550	9:1	27	36	45	63	81	90
	12:1	36	48	60	84	108	120
	15:1	45	60	75	105	135	150
	16:1	48	64	80	112	144	160
	20:1	60	80	100	140	180	200
	21:1	63	84	105	147	189	210
	25:1	75	100	125	175	225	250
	27:1	81	108	135	189	243	270
	28:1	84	112	140	196	252	280
	30:1	90	120	150	210	270	300
	35:1	105	140	175	245	315	350
	36:1	108	144	180	252	324	360
	40:1	120	160	200	280	360	400
	45:1	135	180	225	315	405	450
	49:1	147	196	245	343	441	490
	50:1	150	200	250	350	450	500
	63:1	189	252	315	441	567	630
	70:1	210	280	350	490	630	700
	81:1	243	324	405	567	729	810
	90:1	270	360	450	630	810	900
	100:1	300	400	500	700	900	1000

Motor	Stage 1 & 2	Stage 3					
		3:1	4:1	5:1	7:1	9:1	10:1
BB RS-775	9:1	27	36	45	63	81	90
	12:1	36	48	60	84	108	120
	15:1	45	60	75	105	135	150
	16:1	48	64	80	112	144	160
	20:1	60	80	100	140	180	200
	21:1	63	84	105	147	189	210
	25:1	75	100	125	175	225	250
	27:1	81	108	135	189	243	270
	28:1	84	112	140	196	252	280
	30:1	90	120	150	210	270	300
	35:1	105	140	175	245	315	350
	36:1	108	144	180	252	324	360
	40:1	120	160	200	280	360	400
	45:1	135	180	225	315	405	450
	49:1	147	196	245	343	441	490
	50:1	150	200	250	350	450	500
	63:1	189	252	315	441	567	630
	70:1	210	280	350	490	630	700
	81:1	243	324	405	567	729	810
	90:1	270	360	450	630	810	900
	100:1	300	400	500	700	900	1000
BAG Motor	9:1	27	36	45	63	81	90
	12:1	36	48	60	84	108	120
	15:1	45	60	75	105	135	150
	16:1	48	64	80	112	144	160
	20:1	60	80	100	140	180	200
	21:1	63	84	105	147	189	210
	25:1	75	100	125	175	225	250
	27:1	81	108	135	189	243	270
	28:1	84	112	140	196	252	280
	30:1	90	120	150	210	270	300
	35:1	105	140	175	245	315	350
	36:1	108	144	180	252	324	360
	40:1	120	160	200	280	360	400
	45:1	135	180	225	315	405	450
	49:1	147	196	245	343	441	490
	50:1	150	200	250	350	450	500
	63:1	189	252	315	441	567	630
	70:1	210	280	350	490	630	700
	81:1	243	324	405	567	729	810
	90:1	270	360	450	630	810	900
	100:1	300	400	500	700	900	1000

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VersaPlanetary Load Ratings Guide

Table 1 - 3 Stage VersaPlanetary with 3/8" Hex Output Shaft (Cont.)

Motor	Stage 1 & 2	Stage 3					
		3:1	4:1	5:1	7:1	9:1	10:1
775pro	9:1	27	36	45	63	81	90
	12:1	36	48	60	84	108	120
	15:1	45	60	75	105	135	150
	16:1	48	64	80	112	144	160
	20:1	60	80	100	140	180	200
	21:1	63	84	105	147	189	210
	25:1	75	100	125	175	225	250
	27:1	81	108	135	189	243	270
	28:1	84	112	140	196	252	280
	30:1	90	120	150	210	270	300
	35:1	105	140	175	245	315	350
	36:1	108	144	180	252	324	360
	40:1	120	160	200	280	360	400
	45:1	135	180	225	315	405	450
	49:1	147	196	245	343	441	490
	50:1	150	200	250	350	450	500
	63:1	189	252	315	441	567	630
	70:1	210	280	350	490	630	700
	81:1	243	324	405	567	729	810
	90:1	270	360	450	630	810	900
	100:1	300	400	500	700	900	1000
Mini CIM	9:1	27	36	45	63	81	90
	12:1	36	48	60	84	108	120
	15:1	45	60	75	105	135	150
	16:1	48	64	80	112	144	160
	20:1	60	80	100	140	180	200
	21:1	63	84	105	147	189	210
	25:1	75	100	125	175	225	250
	27:1	81	108	135	189	243	270
	28:1	84	112	140	196	252	280
	30:1	90	120	150	210	270	300
	35:1	105	140	175	245	315	350
	36:1	108	144	180	252	324	360
	40:1	120	160	200	280	360	400
	45:1	135	180	225	315	405	450
	49:1	147	196	245	343	441	490
	50:1	150	200	250	350	450	500
	63:1	189	252	315	441	567	630
	70:1	210	280	350	490	630	700
	81:1	243	324	405	567	729	810
	90:1	270	360	450	630	810	900
	100:1	300	400	500	700	900	1000

Motor	Stage 1 & 2	Stage 3					
		3:1	4:1	5:1	7:1	9:1	10:1
CIM	9:1	DO NOT USE					
	12:1						
	15:1						
	16:1						
	20:1						
	21:1						
	25:1						
	27:1						
	28:1						
	30:1						
	35:1						
	36:1						
	40:1						
	45:1						
	49:1						
	50:1						
	63:1						
	70:1						
	81:1						
	90:1						
	100:1						
NEO	9:1	DO NOT USE					
	12:1						
	15:1						
	16:1						
	20:1						
	21:1						
	25:1						
	27:1						
	28:1						
	30:1						
	35:1						
	36:1						
	40:1						
	45:1						
	49:1						
	50:1						
	63:1						
	70:1						
	81:1						
	90:1						
	100:1						

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Table 1 - 3 Stage VersaPlanetary with 3/8" Hex Output Shaft (Cont.)

		Stage 3					
Motor	**Stage 1 & 2**	3:1	4:1	5:1	7:1	9:1	10:1
Falcon 500	9:1	DO NOT USE					
	12:1						
	15:1						
	16:1						
	20:1						
	21:1						
	25:1						
	27:1						
	28:1						
	30:1						
	35:1						
	36:1						
	40:1						
	45:1						
	49:1						
	50:1						
	63:1						
	70:1						
	81:1						
	90:1						
	100:1						

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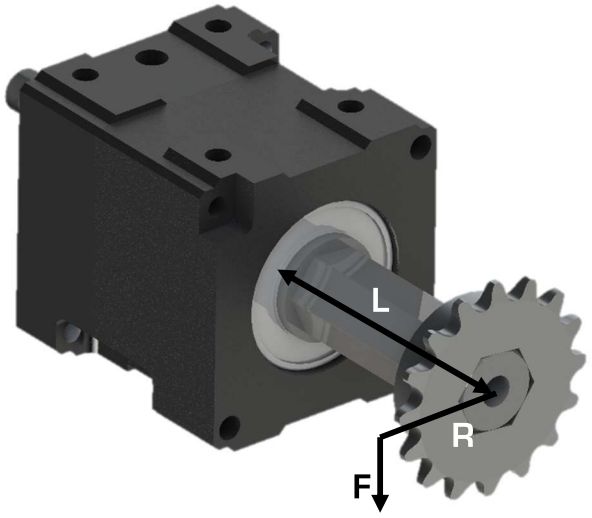
Combined Load Ratings Tables

The VersaPlanetary gearbox is designed to accommodate significant overhung loading with an unsupported shaft. However, the torque carrying capability significantly increases if the gear, sprocket, or pulley is placed close to the mounting face or the end of the shaft is supported with another bearing.

The VersaPlanetary gearbox may fail in one of two different modes: (1) output shaft yield stress failure or (2) 7:1, 9:1, or 10:1 carrier plate ultimate stress failure. However, improper lubrication combined with typical FRC practice robot driving time would also result in eventual gear failure. Refer to the "VersaPlanetary User's Guide" for more information on proper maintenance.

Failure mode #2 (carrier plate failure) is not dependent upon the overhung loading and is solely determined by Table 9. However, all other failure modes listed in Table 9 are beam yield failures and that depend on both torsional failure and beam bending failure.

Failure mode #1 depends upon distance from the mounting face, gear/sprocket diameter, and torque load. Users should reference Table 9 and the below charts to determine the maximum allowable loading for your design.



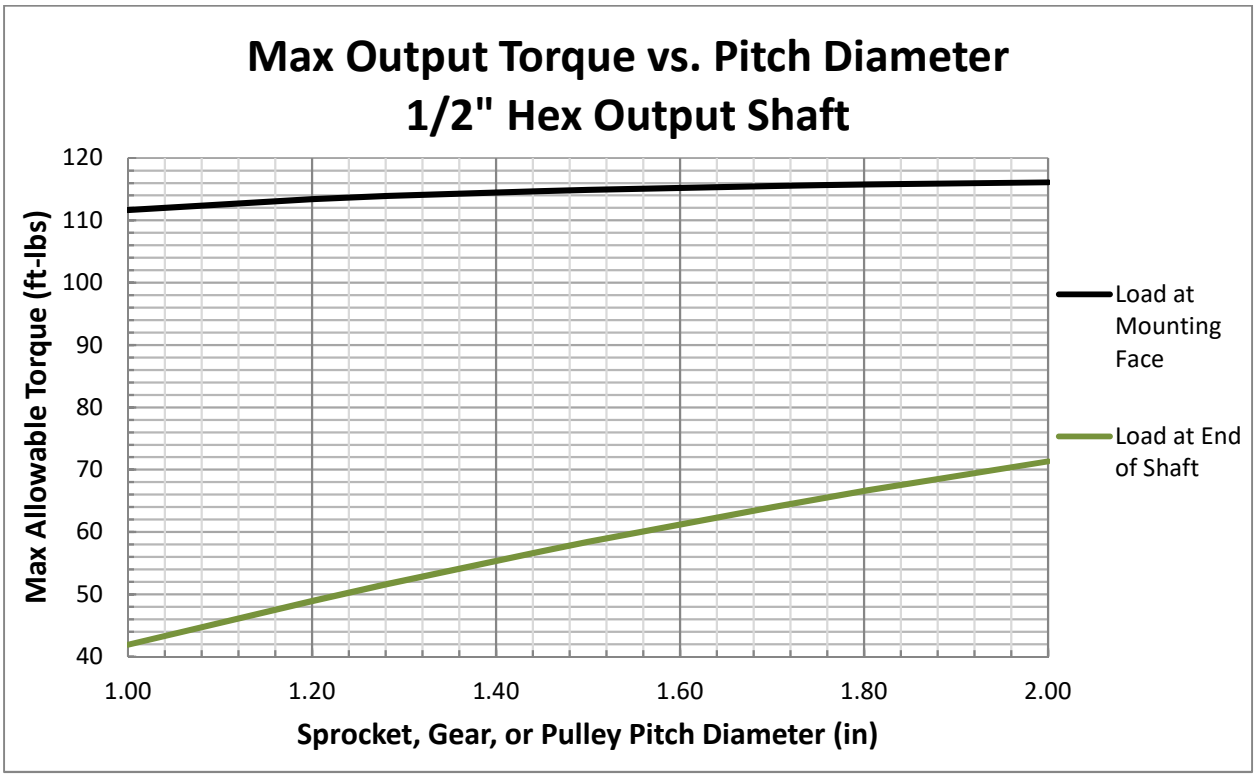
VersaPlanetary v2 Shown

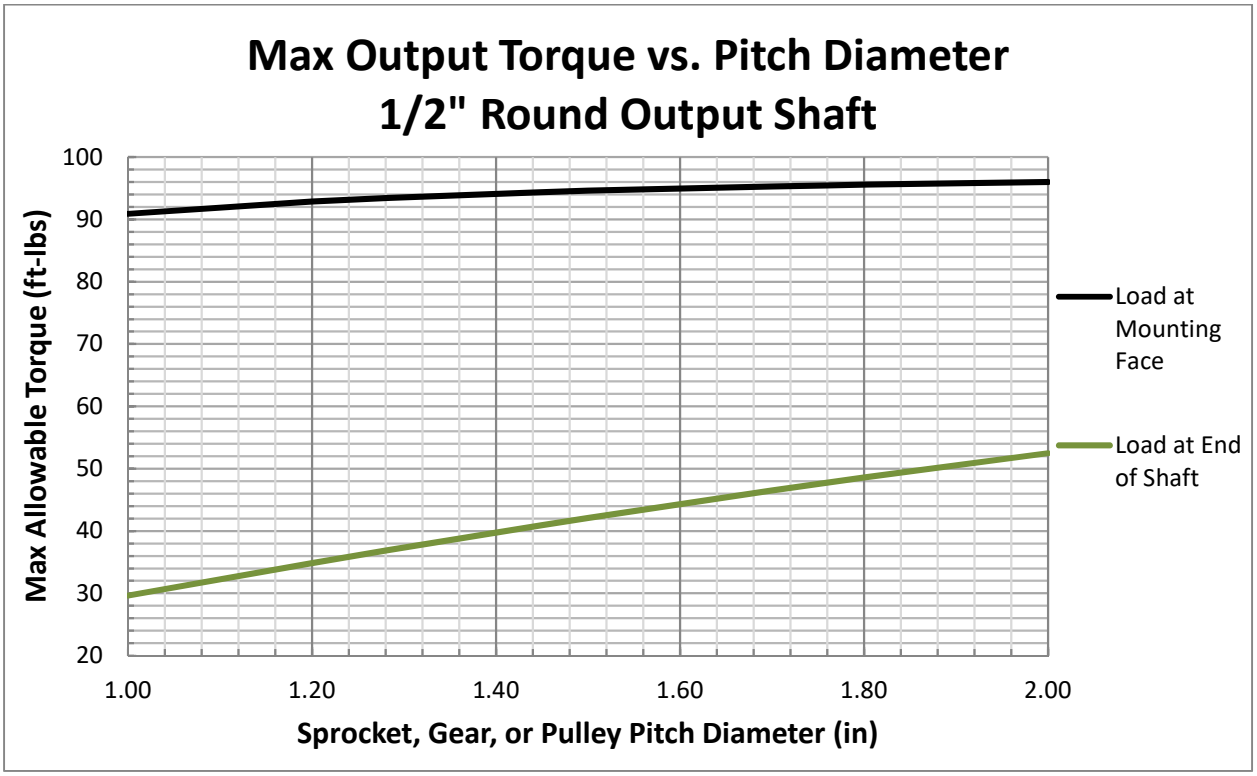
Failure Mode	N-m	ft-lbs
1/2" Hex Shaft Yield Stress	157	116
1/2" Round Shaft Yield Stress	130	96
3/8" Hex Shaft Yield Stress	57	42
CIM Motor Shaft Yield Stress	29	21
7:1, 9:1, 10:1 Carrier Plate Yield Stress	100	74

Table 9 – VersaPlanetary Output Torque Limits (Pure Torsion)

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